

LAB 04 ANALYSIS

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Instructions

1. Update the README.md file and the test_lab04.s documents in the Lab04 folder with the documents listed in Week 15 of Brightspace.
2. Run `make sim_lab04` from the repo root.
3. Open `dump.vcd` in Surfer and observe the pipeline stages.
4. Use the waveforms to answer each question below.

Surfer Setup Walkthrough

The names below are the **actual signal names** you should search for in Surfer.

Register Signal	Meaning
test_Educore.educore.register_file.rX00	X0
test_Educore.educore.register_file.rX01	X1
test_Educore.educore.register_file.rX02	X2
test_Educore.educore.register_file.rX03	X3
test_Educore.educore.register_file.rX04	X4
test_Educore.educore.register_file.rX05	X5
test_Educore.educore.register_file.rX06	X6
test_Educore.educore.register_file.rX07	X7



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test_Educore.educore.register_file.rX09	X9
test_Educore.educore.register_file.rX10	X10
test_Educore.educore.register_file.rX11	X11
test_Educore.educore.register_file.rX12	X12

Part A: Register Values

Step #1:

In the terminal run the code `make sim_lab04`

Step #2:

Open the dump.vcd file in Surfer. Then fill out the table below:

Register	Expected Value	Your Observed Value	Match? (Y/N)
X0	0xF (15)	15	Y
X1	0xE (14)	14	Y
X2	0xD (13)	13	Y
X3	0xC (12)	12	Y
X4	0xB (11)	11	Y
X5	0x10 (16)	16	Y
X6	0x1B (27)	27	Y



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X7	0x1 (1)	1	Y
X9	0x1B (27)	27	Y
X10	likely UNDEF due to RAW hazard	UNDEF	Y
X11	likely UNDEF due to RAW hazard	UNDEF	Y
X12	likely UNDEF due to RAW hazard	UNDEF	Y

Part B: Adding two NOP commands between ADD x9, x1, x2 and ADD x10, x9, x3

Step #1:

In the test_lab04.s replace the code only within Part 3 to the following:

```
_test2:
    ADD    X9, X1, X2          // X9 = X1 + X2 = 27
    NOP                                // first NOP command
    NOP                                // second NOP command
    AND    X10, X9, X3         // X10 = X9 AND X3
    ORR    X11, X5, X9         // X11 = X5 OR X9
    SUB    X12, X9, X7         // X12 = X9 - X7
```

Step #2:



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In the terminal rerun the code `make sim_lab04`

Step #3:

Open the dump.vcd file in Surfer. Then fill out the table below:

Register	Expected Value	Your Observed Value	Match? (Y/N)
X9	0x1B (27)	27	Y
X10	likely UNDEF due to RAW hazard	UNDEF	Y
X11	x5 = 16 or x9 = 27 Which one?	27	Y
X12	0x1A (26)	26	Y

1. Why is x10 still UNDEF while x11 and x12 completed?

x10 is still UNDEF because AND executes one cycle before ADD writes back the updated value for x9. Since x9 hasn't been updated yet, the old value is used. The reason x11 and x12 complete is because, by the time their instructions reach the stage where they read x9, ADD has already finished writing back the updated value.

If you find it helpful for answering this question you may want to add these signals to Surfer:

Signal	What It Shows
test_Educore.educore.clk	Clock cycles
test_Educore.educore.PC	Program Counter in Fetch
test_Educore.educore.ID_PC	Program Counter for Decode-stage



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	instruction
test_Educore.educore.instruction_memory_v	Instruction currently being fetched from instruction memory
test_Educore.educore.instruction	Instruction currently being decoded
test_Educore.educore.EX_exec_n	One Execute-stage operand
test_Educore.educore.EX_exec_m	Second Execute-stage operand or immediate
test_Educore.educore.alu_out	ALU result in Execute
test_Educore.educore.WB_write_en	Whether Writeback is writing a register this cycle
test_Educore.educore.WB_rd_addr	Which register number is written in Writeback
test_Educore.educore.WB_ex_out	Value being written back

Part C: Adding an additional NOP command between ADD x9, x1, x2 and ADD x10, x9, x3

Step #1:

In the test_lab04.s replace the code only within Part 3 to the following:

```
_test2:
    ADD    X9, X1, X2          // X9 = X1 + X2 = 27
```



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```

NOP                                // first NOP command
NOP                                // second NOP command
NOP                                // third NOP command
AND      X10, X9, X3               // X10 = X9 AND X3
ORR      X11, X5, X9               // X11 = X5 OR X9
SUB      X12, X9, X7               // X12 = X9 - X7

```

Step #2:

In the terminal rerun the code `make sim_lab04`

Step #3:

Open the dump.vcd file in Surfer. Then fill out the table below:

Register	Expected Value	Your Observed Value	Match? (Y/N)
X10	0x1B (27)	8	N

1. Explain why adding the third NOP allowed the register x10 to be updated with a value?

Adding the third NOP stalls the AND instruction long enough for ADD to write back the updated value for x9. Therefore, when AND reaches the point where it needs x9, it is able to read the updated value.

Push Your Work to GitHub

After you finish the worksheet and any required edits in ``test_lab04.s``, save



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your work and push it to the repo:

```
git add .  
git commit -m "Complete Lab 04"  
git push
```



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